

Safety Procedures — Tide Motors Assembly Plant 7

Sterling Heights, MI · Document ID: TM7-SAF-001 · Revision 3 · Effective 01 Feb 2026

Document ID	TM7-SAF-001	Revision	3
Plant	Tide Motors Assembly Plant 7	Effective Date	01 Feb 2026
Document Owner	Plant Safety Manager: G. Okafor	Classification	Controlled — Site Use

1. Purpose & Scope

This procedure governs personnel safety at Tide Motors Assembly Plant 7 (Sterling Heights, MI) covering Body-in-White, Paint, General Assembly, and Battery Pack Assembly. Aligns with IATF 16949 clause 6.1.2.3, OSHA General Industry standards, and the Tide Global Safety Standard TGS-08.

2. Regulatory & Standards Basis

Applicable: IATF 16949, OSHA 29 CFR 1910, ISO 14001.

3. Site Hazard Inventory

The principal site hazards at Tide Motors Assembly Plant 7 (Sterling Heights, MI) include: robotic work envelopes, weld arc/fume, pinch points, lithium battery assembly. The full hazard inventory is summarized below; refer to the linked SOPs for detailed controls.

ID	Hazard	Primary Control
H-01	Robotic work envelopes (BIW)	Light curtains and area scanners; entry requires LOTO and key transfer.
H-02	Weld arc/fume	Hood ventilation; auto-darkening helmet; chromium VI monitoring quarterly.
H-03	Pinch points (conveyor, jigs)	Two-hand controls; e-stop pull cords at 10 ft intervals.
H-04	Lithium-ion battery thermal runaway	Pyro-bay isolation; Class D extinguisher; LFP cell screening.
H-05	Forklift and AGV traffic	Marked pedestrian zones; AGVs limited to 1.5 m/s in mixed areas.

4. Procedures

4.1 Robot Cell Entry

Entry permitted only via interlocked gate after Robot Programmer transfers safety key. Teach pendant in T1 mode (250 mm/s) during any in-cell work. Reference: SAF-TM7-RC-03.

4.2 Lockout/Tagout

Multi-energy machines require zero-energy verification across electrical, hydraulic, and pneumatic circuits. Stored hydraulic pressure bled to gauge zero with 5-minute hold.

4.3 Battery Incident Response

Smoke or off-gassing from a pack triggers immediate area evacuation to a 15 m radius. Affected pack moved to outdoor quench bay by trained Battery Response Team only. Cell-level thermal probe data preserved for engineering review.

4.4 Ergonomics

Stations with cycle time below 60 seconds undergo annual RULA assessment. Engineering controls preferred over administrative rotation.

5. Roles & Responsibilities

The Plant Safety Manager owns this procedure. Line Supervisors are accountable for daily compliance. Employees and contractors are responsible for stopping work in the event of an unsafe condition (Stop Work Authority).

6. References

- IATF 16949 Section 6.1.2.3
- OSHA 29 CFR 1910.147 — LOTO
- ANSI/RIA R15.06 — Robot Safety
- NFPA 855 — Stationary ESS (battery handling reference)
- Internal: SAF-TM7-RC-03, TGS-08

7. Revision History

Rev	Date	Author	Summary
1	12 Mar 2024	G. Okafor	Initial issue.
2	08 Aug 2025	G. Okafor	Added confined-space monitoring requirements.
3	01 Feb 2026	G. Okafor	Annual review; aligned with updated regulator guidance.

Approvals

Role	Name	Signature	Date
Plant Safety Manager	G. Okafor	_____	01 Feb 2026
Quality Assurance	H. Tanaka, Quality Director	_____	01 Feb 2026
Operations / Site Lead	E. Brennan, Plant Director	_____	01 Feb 2026

CONFIDENTIAL — Controlled Document. Uncontrolled when printed. Reproduction outside the parent organization requires written authorization.